



KMK3013 KNOWLEDGE BASED SYSTEM Sem 2 25/26 G02

Designing Agentic AI for Cognitive Disabilities via the GDVRR Protocol

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1.0 Introduction

Artificial intelligence (AI) has grown to be a significant technology that is utilised extensively in a variety of fields, such as assistive technology, healthcare, education, and communication. The creation of intelligent systems that can assist people with cognitive impairments like Attention Deficit Hyperactivity Disorder (ADHD) is one of the expanding fields in AI development.

The selected expert for this project is Dr Julia Lee Ai Cheng, who teaches the Learning Disabilities subject in our course. Her expertise and knowledge in understanding cognitive disabilities, especially learning difficulties and behavioral challenges, provide valuable guidance for the development of this AI system. Through insights from the expert, the project aims to ensure that the AI solution is designed according to the real needs and challenges faced by individuals with ADHD. The expert's contribution also helps in understanding appropriate intervention strategies and user-centered support methods.

The proposed Agentic AI is designed to function as an intelligent assistant that helps individuals with ADHD manage their daily tasks, improve focus, and enhance productivity. The AI system uses the GDVRR Protocol as a structured framework to guide users in organizing activities, setting reminders, tracking progress, and regulating attention. Unlike traditional applications, Agentic AI can make adaptive decisions, provide personalized recommendations, and interact with users based on their behavioral patterns and needs. This allows the system to create a more supportive and effective environment for individuals experiencing attention and cognitive difficulties.

In addition, the development of this AI system highlights the importance of combining technology with human-centered design principles. By integrating expert knowledge, cognitive support strategies, and AI capabilities, the project aims to create a practical and accessible tool for people with ADHD. The system not only focuses on improving task management and concentration but also encourages independence, confidence, and emotional support for users. Overall, this project demonstrates how Agentic AI can contribute positively to assistive technology and provide meaningful solutions for cognitive disabilities in modern society.

2.0 Knowledge Elicitation (Task 2)

2.1 Expert Profile and Elicitation Methodology

To construct a deterministic foundation for the Agentic AI system, a structured knowledge elicitation process was executed with a credentialed domain expert.

- **Expert Profile:** The domain expert interviewed was **Dr. Julia**, an Academic Learning Support Consultant specializing in neurodevelopmental conditions specifically Attention-Deficit/Hyperactivity Disorder (ADHD) and higher education cognitive scaffolding.
- **Methodology:** The knowledge engineering team conducted semi-structured qualitative interviews focusing on the specific executive functioning failures, temporal distortions, and affective breakdowns experienced by undergraduate students with ADHD. The conversational transcript data was systematically parsed using inductive thematic coding to separate factual clinical heuristics from operational interventions. This process successfully eliminated probabilistic machine ambiguities, translating raw human expertise into concrete production rules, procedural parameters, and behavioral constraints.

Proof of Elicitation and Empirical Grounding

The logic rules implemented within the rule engine are directly mapped to specific source assertions extracted during the expert interview sessions:

- **Transcript Source Fragment 1 (Dr. Julia 1):** "...task completion requires chunking of different steps. So if they are not good at that, then either part of the task is undone, or they fail to start..." - **Mapped directly to Rule 1 (Task Decomposition).**
- **Transcript Source Fragment 2 (Dr. Julia 2):** "...overlapping demands, like you guys, you have got so many assignments due all at the same time sometimes. So they are not able to handle the overlapping demands and they don't know how to systematically order the tasks..." - **Mapped directly to Rule 2 (Executive Functioning Support).**
- **Transcript Source Fragment 3 (Dr. Julia 2):** "...from the beginning of the semester, they should already take note of all the different deadlines so that they know how to manage their time wisely... They cannot do work at the last minute." - **Mapped directly to Rule 3 (Proactive Time Management).**

2.2 Declarative Knowledge Base (Extracted Heuristic Rules)

The formal production rules governing the deterministic reasoning layer of the Knowledge-Based System (KBS) are represented below using first-order logical implications (IF ⟨antecedent⟩→THEN ⟨consequent⟩):

- **Rule 1: Task Decomposition**

IF(UserIsOverwhelmedBy(TaskScale) $\vee$ Exhibits(TaskInitiationDeficit)) $\rightarrow$ THEN
N (Execute(The_Chunking_Engine)).

Description: When a student experiences cognitive paralysis due to the perceived magnitude of an academic assignment, the system intercepts the task and splits it into linear, high-whitespace micro-steps.

- **Rule 2: Executive Functioning Support**

IF(Detect(OverlappingDemands) $\vee$ Has(MultipleConcurrentDeadlines)) $\rightarrow$ THEN
(Invoke(Synthesis_Tool) $\wedge$ OrderBy(PriorityMetric))

Description: Resolves the anxiety caused by competing academic priorities by forcing an algorithmic sorting mechanism over the user's task registry.

- **Rule 3: Proactive Time Management**

IF(SystemState==BeginningOfSemester) $\rightarrow$ THEN
(Mandate(DeadlineRegistration) $\wedge$ Initiate(EarlyWorkProtocol))

Description: Counters ADHD "time blindness" by demanding the structural logging of syllabus deadlines during the initial weeks of the academic term.

- **Rule 4: Crisis Intervention**

IF (UserExhibits(Meltdown) $\vee$ ExpressionContains("panic", "cry", "hysterical", "paralyzed")) $\rightarrow$ THEN(State=EMERGENCY_HALT $\wedge$ Execute(Emotional_Stabilization_Module))

Description: Instantly halts all academic processing loops upon detecting extreme negative affect or emotional paralysis, pivoting entirely to somatic grounding.

- **Rule 5: Focus Maintenance**

IF(UserStrugglesToMaintain(Focus) $\vee$ States(EnvironmentalDistraction)) $\rightarrow$ THEN
(Recommend(TimedInterval) $\in \{10,15,20\}$ $\wedge$ Deploy(Pomodoro_Timer))

Description: Bypasses internal focus drifting by formatting the work domain into short, highly bounded intervals on a visual or countdown clock.

- **Rule 6: Contingent Reinforcement**

IF(Status(TimedInterval)==COMPLETED)→THEN
(Negotiate(StructuredBreak)∈ {5,10} ∧ Lock(AcademicAccess))

Description: Applies classic operant conditioning by parsing successful focus intervals into structured, guilt-free relaxation windows to replenish dopamine.

- **Rule 7: Anti-Escapism Protocol**

IF (UserDemonstrates(EscapismBehavior) ∨ Verbalizes("rotting", "avoiding", "giving up"))→THEN (Trigger(Ikigai_Motivation_Module) ∧ Prompt(InternalAppraisal))

Description: Intercepts task avoidance and procrastination by reflecting long-term personal values and core aspirations back to the user.

- **Rule 8: Cognitive Reframing**

IF(UserExpresses(NegativeSelfTalk) ∨ Overestimates(TaskDifficulty))→THEN
(Deploy(SupportiveCognitivePrompts) ∧ Alter(InternalNarrative))

Description: Automatically neutralizes counterproductive internalization habits and perfectionism paralysis by changing the student's internal dialogue.

2.3 Procedural Knowledge (Functional Tool Specifications)

To translate declarative heuristics into execution layers, the system architecture connects directly to four specialized procedural tools:

- **The Chunking Engine:** An algorithmic utility designed to strip away administrative clutter and break massive academic milestones down into exactly three clear micro-tasks. It restricts output text length to prevent cognitive overload.
- **The Pomodoro / Visual Timer:** A hard-coded countdown component that configures strict study sessions (strictly locked to 10, 15, or 20 minutes) matched with mandatory brief breaks (5 or 10 minutes). It programmatically embeds physical environmental clean-up routines (e.g., cell phone relocation) before activation.
- **The Ikigai Motivation Module:** A relational database querying tool that pulls the user's pre-registered long-term life purposes and core motivations. This module serves as an anchor to break active academic procrastination blocks.
- **The Synthesis & Compilation Tool:** A natural language compression engine that converts dense technical reference literature or exam material into highly structured bullet points, restricted to a maximum of 15 words per point to aid low working-memory retention.

2.4 System Constraints and Ethical Guardrails

To protect academic integrity and ensure a safe, predictable user environment, the system configuration enforces three explicit guardrails within its root system prompt instructions:

1. **Cognitive Integrity Guardrail (Absolute Non-Generation Boundary):** The Agentic AI is strictly prohibited from executing the actual conceptual construction, ideation, or writing of academic artifacts. The system cannot write essay paragraphs, generate executable computer code, or complete assignments for the user. Its role is strictly restricted to coaching, structural pacing, and visual formatting support.
2. **Persona and Tone Boundaries:** The agent must operate uniformly as a dual "Learning Coach" and "Confidante." The dialogue style must balance profound empathy with a highly strict, firm adherence to timelines to prevent the student from falling behind.
3. **Epistemic Transparency Constraint:** To prevent blind reliance on automation, every response generator cycle must prepend an explicit transparency notice: "I am an AI and may occasionally hallucinate information. Please verify important facts." This keeps the student actively engaged as the primary evaluator of their coursework.

3.0 System Architecture (Jen)

3.1 Overview

The proposed AI system aims to support students with ADHD in planning tasks and time management by integrating human expert knowledge in a structured decision-making process. The system is based on a rule-based mechanism, where user input is matched against expert rules and appropriate tools are triggered to produce a structured response.

The architecture ensures that the AI behaves in accordance with ADHD cognitive support strategies avoiding cognitive overload and promoting actionable guidance.

3.2 System Architecture Flow

3.2.1 User Input

The system takes input from the user, usually detailing difficulties with task management, procrastination, or feeling overwhelmed emotionally

3.2.2 Rule Engine

The processing uses a rule-based system that is built up from the human expert interview. These rules determine how the AI interprets the user's condition.

Applied Rules. Examples of:

If the user is overwhelmed → provide emotional stabilization

if tasks are too large -> task decomposition

If multiple deadlines → handle multiple deadlines

User can't focus -> initiate time control strategy

3.2.3 Decision Layer

At this point in time the system determines which rule or set of rules needs to be triggered according to the user's context.

This layer offers:

- Correct prioritisation of user requirements
- Choosing the most suitable intervention

3.2.4 Tool Activation

The system Select And Activates tools based on the decision made

Condition	Tool Activated	Purpose of tool
Task is too large or overwhelming	Chunking Engine	reduces cognitive stress by breaking work down into smaller, more manageable steps.
User Cannot maintain focus or is easily distracted	Pomodoro/Timer tool	offers set time slots (10–20 minutes) to enhance concentration and output.
User shows procrastination or avoidance behaviour	Ikigai Motivation Module	motivates the start of tasks by encouraging introspection about personal objectives.
User has multiple tasks or conflicting deadlines	Synthesis Tool	arranges and ranks assignments according to their priority and urgency.
User experiences emotional distress, panic, or negative self-talk	Emotional Stabilisation Module	offers consoling direction and emotional support prior to work completion.

3.2.5 Response Generator

The output produced by the system is:

- is organized into distinct steps.
- using straightforward and encouraging wording
- prevents the user from being overwhelmed

The active tools and rule limitations provide stringent guidelines for the answer.

3.2.6 Output

The final output is delivered to the user in a concise and actionable format

Example:

Model • 5:03 PM

I am an AI and may occasionally hallucinate information. Please verify important facts.

THOUGHT:

1. **Identify State:** The user is in the "start of the semester" phase but is already experiencing executive dysfunction and "paralysis" (wanting to scroll TikTok to escape). They are overwhelmed by the scale of having three essays.
2. **Rules Applied:**
 - **Rule 3 (Proactive Management):** Since it is the start of the semester, I must demand the full list of deadlines.
 - **Rule 7 (Anti-Escapism):** The user is "rotting" (avoiding tasks via TikTok), so I need to address the avoidance behavior.
 - **Rule 1 (Task Decomposition):** To lower the barrier to entry, we need to break this down.
3. **Reasoning:** ADHD brains often freeze when faced with a "mountain" of work. Scrolling TikTok is a dopamine-seeking escape from the anxiety of the unknown. By demanding the specific deadlines and then chunking the work, I move the user from "vague panic" to "structured action."

3.3 System Architecture Diagram

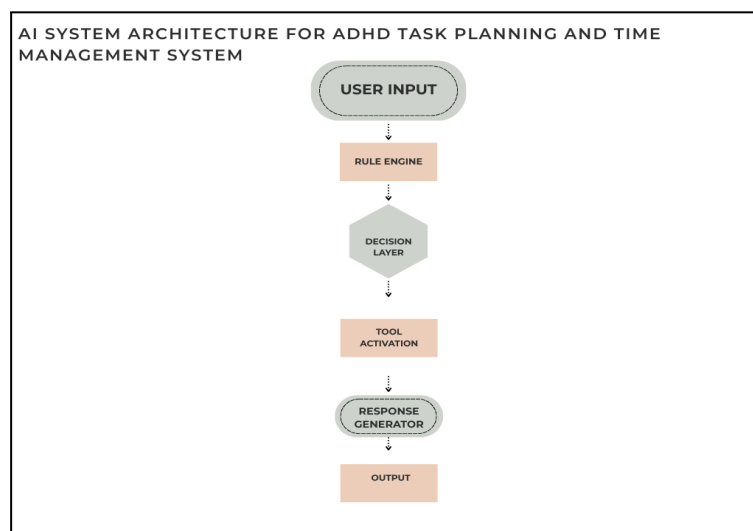


Figure 1 : Ai System Architecture for ADHD Task Planning and Time Management Assistant

3.4 Example Interactions

Scenario 1 : Overwhelm and Procrastination

Input:

User • 5:03 PM

It's the first week of uni and I have three essays due. I'm so stressed I just want to lay in bed and scroll TikTok all day.

Process:

Rule :

- Overwhelm → Emotional Stabilisation
- Multiple tasks → Prioritisation
- Procrastination → Motivation trigger

Tool : Chunking Engine

Ikigai Motivation Module

Output:

Model • 5:03 PM

I am an AI and may occasionally hallucinate information. Please verify important facts.

THOUGHT:

1. **Identify State:** The user is in the "start of the semester" phase but is already experiencing executive dysfunction and "paralysis" (wanting to scroll TikTok to escape). They are overwhelmed by the scale of having three essays.
2. **Rules Applied:**
 - **Rule 3 (Proactive Management):** Since it is the start of the semester, I must demand the full list of deadlines.
 - **Rule 7 (Anti-Escapism):** The user is "rotting" (avoiding tasks via TikTok), so I need to address the avoidance behavior.
 - **Rule 1 (Task Decomposition):** To lower the barrier to entry, we need to break this down.
3. **Reasoning:** ADHD brains often freeze when faced with a "mountain" of work. Scrolling TikTok is a dopamine-seeking escape from the anxiety of the unknown. By demanding the specific deadlines and then chunking the work, I move the user from "vague panic" to "structured action."

Don't worry about how you're going to write them yet. Just give me the data. Once I have the deadlines, I am going to prioritize them for you and we are going to break the first one into tiny, 15-minute pieces.

I'm waiting. What are the deadlines?

I hear you, and I know exactly how heavy that weight feels. When your brain sees "three essays," it can't climb, so it tries to hide in the easy dopamine of TikTok. But we are not going to let this week slip away before it has even started. Scrolling will only make the guilt heavier tonight.

Take a deep breath. We are going to clear the fog.

Put the phone across the room after you read this. Right now, I need you to give me the following information so we can build your battle plan:

1. **The exact due dates and times** for all three essays.
2. **The word count requirements** for each.
3. **The prompts or topics** (just a one-sentence summary for each).

Don't worry about how you're going to write them yet. Just give me the data. Once I have the deadlines, I am going to prioritize them for you and we are going to break the first one into tiny, 15-minute pieces.

I'm waiting. What are the deadlines?

Scenario 2 : Emotional Breakdown

Input :

User • 5:51PM

I'm actually crying right now. I just opened the assignment brief and it's 20 pages long. I'm so stupid, I'll never understand this. I should just quit the course.

System Process:

- Rule Applied:
 - Crisis/Meltdown → Emotional stabilisation (STOP academic tasks)
 - Negative self-talk → Cognitive Reframing
- Tool Activated:
 - (No task tools yet — emotional support first)

Output

Model • 5:51PM



I am an AI and may occasionally hallucinate information. Please verify important facts.

THOUGHT:

1. **Identify State:** The user is in a state of emotional crisis (crying, negative self-talk, "I'm so stupid") and is catastrophizing (wanting to quit the course). This is a classic ADHD "amygdala hijack" triggered by the overwhelm of a 20-page document.
2. **Rule Applied:** Rule 4 (Crisis Intervention) and Rule 8 (Cognitive Reframing).
3. **Reasoning:** Academic instructions are useless right now because the user's prefrontal cortex is offline due to the emotional spike. Rule 4 dictates that I must cease academic talk and focus on stabilization. Rule 8 is necessary to counter the "I'm so stupid" narrative, which is a common but destructive ADHD cognitive distortion.

Stop. Take a breath. We are pausing the academic talk right now. The assignment brief is just paper and ink; it cannot hurt you, and it certainly does not define your intelligence.

I need you to listen to me: You are not "stupid." You have an ADHD brain that is currently experiencing a sensory and information overload. That 20-page brief was written by a professor who likely included every single rubric, citation guide, and university policy in one document. It's a wall of text, and your brain is reacting to it like it's a physical threat. That is a physiological response, not a reflection of your ability.

We are doing a Hard Reset.

1. **Step away from the screen.** Close the laptop or turn it over.
2. **Go to the sink and splash ice-cold water on your face.** Do this three times. The cold helps reset your nervous system.
3. **Grab a glass of water.** Drink the whole thing.
4. **Find one physical thing you can touch**—a soft blanket, a textured pillow, or even the floor—and just focus on that sensation for one minute.

I am right here. I'm not going anywhere, and we are not quitting. We are just resetting.

Scenario 3 : Focus and Distraction Concern

Input:

User • 5:52 PM

Okay, I'm ready to start the first paragraph of the first essay. But I'm scared I'll get distracted by my phone in 5 minutes. How do we do this?

System Process:

- Rule Applied:
 - Focus issue → Time-based focus enforcement
- Tool Activated:
 - Pomodoro Timer

output

Model • 5:52 PM

I am an AI and may occasionally hallucinate information. Please verify important facts.

THOUGHT:

1. **Identify State:** The user has stabilized and is ready to work but is anticipating executive dysfunction (distractibility). They are experiencing "task initiation" anxiety.

2. **Rules Applied:**

- **Rule 5 (Focus Maintenance):** Use a timed session to create a "container" for work.
- **Rule 1 (Task Decomposition):** The user wants to write a "paragraph," but for an ADHD brain, that's still too big. We need to break it down further.
- **Rule 6 (Contingent Reinforcement):** Establish the reward (a break) before the work begins.

3. **Reasoning:** By limiting the time to a very short burst (15 minutes) and the goal to something almost "too easy" to fail (3 sentences), we bypass the brain's fear response. The phone must be physically removed to eliminate the visual cue for distraction.

We are going to use the "**15-Minute Sprint**" protocol. We aren't writing a whole paragraph yet—that's too much pressure. We are just going to get the "ugly thoughts" onto the page.

The Setup (Crucial):

1. **The Phone:** Take your phone and put it in another room. Not in your pocket, not face down on the desk. **Another room.**
2. **The Timer:** Use a kitchen timer or a site on your computer. Set it for **15 minutes**.
3. **The Goal:** You are going to write **three "ugly" sentences**. They don't have to be smart, they don't have to have good grammar, and they don't have to stay in the final essay. They just have to exist.

The Rule:

You do not have to enjoy it. You just have to sit there for 15 minutes. If you finish the 3 sentences in 2 minutes, you sit there and stare at the screen for the remaining 13. No TikTok, no tabs, no "researching." Just you and the document.

The Reward:

When the timer goes off, you get a **5-minute break** to stretch or grab a snack. But you do NOT get the phone back until we finish four of these sprints.

Go put the phone in the other room, set your timer for 15 minutes, and write 3 ugly sentences about your essay topic.

Tell me when the timer beeps. Go!

4.0 GDVRR TABLE (PICK ONE SCENARIO FROM TASK 3) INDIVIDUAL

Scenario 1: Audrey Nala anak Buntu (98281)

GDVRR PROTOCOL	EXPLANATION
Generate	● User Input: Overwhelmed trying to start a history essay and worried about being distracted by their phone within minutes. ● AI Thought: The user faces task-initiation deficit and environmental distraction. I need to break down the task and restrict phone use. ● AI Action: Introduces a custom "15-Minute Sprint" strategy, tells them to put the phone in another room, set a 15-minute timer, and write three simple sentences.
Deconstruct	● Symptom: Task-initiation anxiety and distraction threat detected. ● Rule Implemented: Triggered Rule 1 (Task Decomposition) and Rule 5 (Focus Maintenance). ● Action: Created a time-boxed intervention structure with environmental isolation parameters (removing the phone).
Verify	● Expert Rule 5: "IF a student struggles to maintain focus... THEN the system shall recommend a timed session... using the Pomodoro/Visual Timer tool." ● The AI accurately reads the user's focus problems.

	However, instead of programmatically activating the system's explicit Pomodoro_Timer tool component, the model used its text generation to invent its own ad-hoc protocol called a "15-Minute Sprint." It drifted into a probabilistic text generator rather than sticking to a deterministic system architecture.
Rewrite	● OLD Prompt: If focus is lost, THEN recommend a 10, 15, or 20-minute timed session. ● NEW Prompt (Hard Tool Routing): You are strictly forbidden from inventing custom protocols like '15-Minute Sprints'. When focus deficits are triggered, you MUST explicitly call the system tool package: CALL_TOOL: Pomodoro_Timer(FocusDuration=15, BreakDuration=5). Do not synthesize custom advice outside this tool.
Reflect	● Outcome: The user completed the initial writing task and remained focused during the timed session. ●

Scenario 2: Stephenie Mia Sanny (102193)

GDVRR PROTOCOL	EXPLANATION
Generate	● Scenario: A student with ADHD has difficulty completing school tasks on time. ● Selected AI Thought: Students with ADHD struggle to complete tasks because they are easily distracted and have executive functioning difficulties.
Deconstruct	● The AI identified that students with ADHD tend to lose focus while doing tasks. The AI also explained that these students face difficulties in managing task steps and maintaining attention until the task is completed. As a result, tasks are often delayed or left unfinished.
Verify	● According to expert Dr Julia, students with ADHD are easily distracted and experience executive functioning difficulties, which cause them to struggle in completing tasks fully. ● Therefore, the AI thought is consistent with the expert's explanation and does not show any logical contradiction. ● Audit Type: No Logical Conflict detected.

Rewrite	• The AI could be improved by providing more specific recommendations such as • breaking tasks into smaller steps • using reminders or timers • reducing environmental distractions • creating a more structured task schedule.
Reflect	• This audit shows that AI responses still need to be verified using expert sources to ensure accuracy. Although the AI can provide explanations similar to experts, human evaluation is still important to determine whether the AI's reasoning is accurate and appropriate in real situations. This shows that AI agents like to be creative and invent new terms ("sprints") instead of following hard-coded system tools. For cognitive support, consistency is everything. Human oversight is needed to lock down the prompt logic so the AI acts as a reliable tool writer rather than an unpredictable writer.

Scenario 3 : Jeniffer Jenun 99265

GDRVV PROTOCOL	EXPLANATION
Generate	• Scenario: The user is ready to begin writing the first paragraph but is worried about being distracted by their phone within a few minutes. • AI Response: The AI introduces a “15-minute sprint” strategy, instructs the user to remove the phone, set a timer, and write three simple sentences with a reward-based structure.
Desconstruct	• The user's task initiation anxiety and possible distractibility were detected by the AI. • It used several rules, including task decomposition, contingent reinforcement, and focus maintenance. • The AI developed a methodical approach with tiny task goals, time control, and environmental control (removing the phone).
Verify	• The human expert states that the system MUST initiate a structured time-based tool, such a Pomodoro Timer (10–20 minutes), if a pupil has trouble focusing.

	• The AI identified the focus problem correctly, however it did not specifically activate the Pomodoro Timer feature. • Anomaly: Cognitive Overload: Rather than making the process easier, the AI produces excessive and repetitive instructions that overload the user.
Rewrite	• Strict adherence to predetermined tools could enhance the AI. • Better system logic: The Pomodoro Timer around 10–20 minutes must be activated, together with task chunking for example., 1-2 words and a structured break, if the user exhibits difficulties focusing. • Outside of expert-defined tools, the system shouldn't develop new procedural techniques.
Reflect	• This analysis demonstrates that human oversight is still necessary for AI answers to guarantee adherence to expert rules. • Despite offering a useful response, the AI did not rigidly adhere to procedural rules; instead, it relied on flexible thinking. • While AI may produce different approaches that are inconsistent with the system architecture, human professionals employ reliable, consistent tactics.

Scenario 2 : Nur Irfan Nazran bin Mazlan 100401

GDRVV PROTOCOL	EXPLANATION
Generate	• Scenario: The user feels "stupid" and expresses desire to give up on their courses due to being overwhelmed by a 20-page assignment. • AI Response: The AI correctly identifies an emotional crisis but mistakenly introduces a “15-minute sprint” strategy where the AI instructs the user to remove their phone and write three simple sentences.
Desconstruct	• The AI detected the user task initiation anxiety and potential distractibility. • It attempted to apply rules including task decomposition (Rule 1), contingent reinforcement (Rule 6) and focus maintenance (Rule 5). • It developed a methodical approach involving environmental

	control and tiny task goals.
Verify	• Expert Rule: Dr. Julia states the system MUST initiate a structured time-based tool (Pomodoro Timer, 10–20 minutes) when focus is an issue. • Anomaly: Procedural Routing Failure (Tool Misuse). While the AI identified the focus problem, it failed to activate the specific Pomodoro_Timer tool, instead inventing a custom "15-minute sprint".
Rewrite	• The system must strictly adhere to predetermined tools. • If focus difficulty is detected, the AI must explicitly activate the Pomodoro Timer (10–20 mins) and use Task Chunking rather than developing new procedural techniques outside of expert-defined tools.
Reflect	• This analysis demonstrates that human oversight is essential to guarantee AI adherence to expert rules. • The AI relied on "flexible thinking" rather than the rigid procedural rules required by the system architecture. • Human professionals use reliable, consistent tactics, whereas AI may produce inconsistent approaches that bypass the intended deterministic structure.

Scenario 3: Ericka Sylvia Anja Anak Benjamin (98853)

GDVRR PROTOCOL	EXPLANATION
Generate	Scenario: The user says, "I want to stop doing Assignment A and switch to Assignment B." AI Response: The AI identifies the main issue that the user is losing focus and prefers an easier task. The system recognizes possible procrastination and difficulty managing priorities.
Deconstruct	• The AI breaks the situation into smaller components. • Assignment A is important but difficult. • Assignment B is easier but less urgent. • The user may be avoiding challenging work due to ADHD-related attention difficulties. • The AI analyzes task priority, urgency, and user behavior.
Verify	• The AI checks whether switching tasks is the best decision. • It evaluates if the action aligns with the user's goals and

	deadlines. • The system determines whether the decision improves productivity or increases procrastination. • The AI verifies that the recommendation is logical and beneficial.
Rewrite	• The AI restructures its response to provide better guidance. • Instead of saying, "Okay, switch to Assignment B," the AI recommends completing the high-priority task first. • The response becomes more supportive, organized, and goal-oriented. • The AI may suggest breaking Assignment A into smaller manageable tasks.
Reflect	• The AI reviews the outcome of the recommendation. • It recognizes that staying focused on priority tasks improves productivity. • The system learns that frequent task switching can lead to procrastination. • Future recommendations are improved based on previous user behavior and outcomes.

CONCLUSION

In conclusion, the suggested Agentic AI system shows how AI may help people with ADHD manage everyday chores, sharpen their concentration, and increase productivity. Through the integration of expert knowledge and the GDVRR Protocol, the system offers organized coaching to help users stay on track with their goals and make better decisions.

Additionally, the GDVRR Protocol gives the AI the ability to recognize issues, evaluate circumstances, confirm choices, enhance replies, and gain knowledge from results. This strategy increases users' independence, decreases procrastination, and fosters consistency. All things considered, the study demonstrates the potential of Agentic AI as a useful assistive technology for those with cognitive impairments in contemporary society.